

Day-4: IP PARAMETER

Finding host IP in per N/W:

50. x.x.x
 ↓
 Host

place

Step:1 Separate host bit into separate host values.

50.

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

 .

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

 .

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

50. 0.00000000 00000000 00000000

50. 11111111 11111111 11111111

Starting → 50.0.0.0 \ 8

 ↓
 prefix length [No. of N/w bits]

 ↓
 This is not IP. It is called
 as network ID because all host
 are zero.

Ending → 50.255.255.255

 ↓
 This is called as broadcast IP

50.0.0.0 | 8 → NETWORK ID

50.0.0.1 → FIRST IP

50.0.0.2

↓

50.0.0.255

50.0.1.0

50.0.1.1

↓

50.0.1.255

50.0.2.0

50.0.2.255

↓

50.0.255.255

50.1.0.0

↓

50.255.255.254 → LAST IP

50.255.255.255 → BROADCAST IP

FORMULA:

$$2^{\text{host bit}}$$

2 → Binary (1,0)

No. of host bits:
class-A → 16,777,214
class-B → 65,534
class-C → 254

$$2^{24} = 16,777,216$$

$$- 2 \text{ [Network ID \& Broadcast IP]}$$

$$16,777,214$$

* Network ID are mainly used for permission purposes.

* Broadcast IP are used for data communication

Eg:2 172.18.216.74 [class B]

SOL: 172.18.216.74 → 172.18.x.x

$$\Rightarrow 172.18 \cdot \begin{array}{|c|c|c|c|c|c|c|c|} \hline 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array} \cdot \begin{array}{|c|c|c|c|c|c|c|c|} \hline 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

Network ID

$$\hookrightarrow 172.18.0.0 / 16$$

$$172.18.0.1 \rightarrow 1^{st} \text{ IP}$$

$$172.18.255.254 \rightarrow \text{last IP}$$

Total no. of bits: $2^{16} = 65,536$

$$- 2$$

$$65,534$$

Eg:3 192.192.87.214

SOL: It is class C. 192.192.87.x

$$\Rightarrow 192.192.87 \cdot \begin{array}{|c|c|c|c|c|c|c|c|} \hline 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline \end{array}$$

Network ID

$$\hookrightarrow 192.192.87.0 / 24$$

$$192.192.87.1 \rightarrow 1^{st} \text{ IP}$$

$$192.192.87.254 \rightarrow \text{last IP}$$

$$192.192.87.255$$

Total no. of host bits:

$$2^8 = 256$$

$$- 2$$

$$254$$

→ Broadcast IP.

Subnet Mask:

* It is used to mention Network portion in IP.

Condition: Put binary as 1 for N/W.

Eg: $50.0.0.10 \rightarrow$

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

$\Rightarrow 255.0.0.0.$

$172.16.0.10 \rightarrow$

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

128	64	32	16	8	4	2	1
-----	----	----	----	---	---	---	---

$\rightarrow 255.255.0.0.$

$192.168.1.76$

$\Rightarrow 255.255.255.0.$

ENTHIRAN IP ADDRESS: $108.11.0.1$

$108.x.x.x$

$108.0.0.0 \rightarrow$ NETWORK ID

$108.0.0.1 \rightarrow$ FIRST IP

$\vdots$

$108.255.255.254 \rightarrow$ LAST IP

$108.255.255.255 \rightarrow$ BROADCAST - IP

$108.0.0.0 \mid 8 \rightarrow$ PREFIX LENGTH

$255.0.0.0 \rightarrow$ SUBNET MASK